MReactor: Offline Multiple Appropriate Facial Reaction Generation with Hierarchical Cognitive Disentanglement

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Abstract

Facial reactions are central to dyadic communication, revealing how listeners perceive and respond to speakers' behaviours. Unlike traditional facial reaction generation methods that assume a one-to-one mapping, human reactions are inherently one-to-many, motivating the Multiple Appropriate Facial Reaction Generation (MAFRG) task. Despite the significant advances achieved in recent years, existing approaches still face challenges in noisy multi-modal perception, entangled local and global dynamics, and limited semantic controllability. To tackle these problems, we present MReactor *in this work, which is the novel offline and cognitively grounded MAFRG framework inspired by the Human Model Processor. Specifically, the* MReactor *integrates (i) multi-modal behaviour perception for stable cue extraction, (ii) hierarchical cognition disentanglement for fine- and coarse-grained cues, and (iii) multiple appropriate generation for diverse and contextually coherent reactions via contextual behaviour and recovery expression. Extensive experiments on the MARS dataset show that our* MReactor *achieved state-of-the-art results in synchrony, appropriateness and diversity, producing natural and human-like facial reactions.*

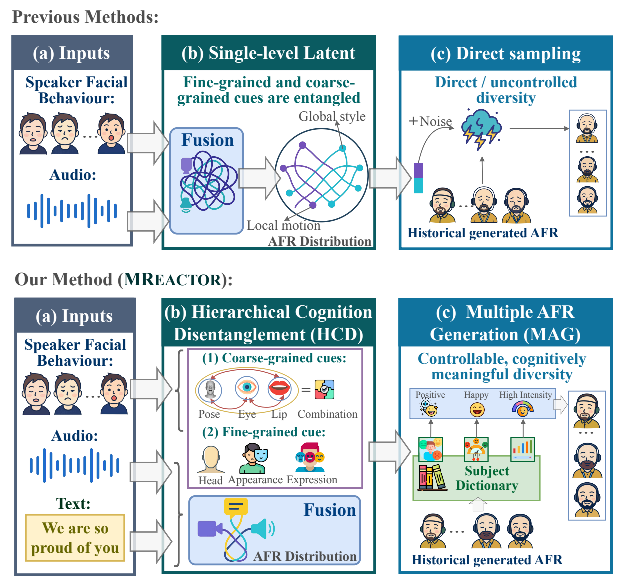


Figure 1. Comparison of our MReactor with existing approaches. We include prior **uni-modal** methods such as GeneFace [44], StyleCLIP [23], and DyadGAN [8]; **bimodal** methods including Wav2Lip [27], BEAMER [7], REGNN [43], ReactFace [13], DreamTalk [15], and PerFRDiff [53]; and our multi-modal **MReactor** method.

1. Introduction

Facial reactions in human-human dyadic interactions refer to non-verbal facial behaviour that arise in response to the behaviour expressed by the conversational partner, which are integral to interpersonal communication, offering vital cues for understanding underlying intentions and emotional dynamics [17]. Generating contextually appropriate facial reactions is essential for improving the naturalness and social effectiveness of human–computer interaction. In settings such as virtual customer support and digital companions, adaptive facial feedback enhances user engagement and perceived empathy, highlighting the need for reliable automatic facial reaction generation (FRG) systems.

Early efforts on FRG have primarily focused on generating responsive, synchronised facial expressions [6, 19, 20, 30, 44, 49] and head movements [51] from the conversational partner's behaviour. However, they typically treat FRG as a deterministic *one-to-one mapping* task, overlooking its inherent *one-to-many mapping* nature arising from the variability and uncertainty of human conversational behaviour reactions [34]. To explicitly address this variability, Multiple Appropriate Facial Reaction Generation (MAFRG) [31, 32, 34] has emerged to produce multiple diverse yet contextually appropriate facial reactions (AFRs) from each speaker behaviour. Recent MAFRG methods have explored various approaches: REGNN [43] employs reversible graph neural networks for distribution learning, PerFRDiff [53] leverages personalized diffusion models, PerReactor [52] introduces personalized weight editing, and SC-Diff [16] utilities scattering-conditioned diffusion. While these advances have improved diversity and appropriateness, the MAFRG task fundamentally spans three cognitive levels—perceiving, interpreting, and acting—revealing two unresolved problems as discussed below.

Problem I: *Hierarchical Speaker Behaviour Interpretation.* Most existing approaches encode speaker behaviour cues using single-level latent representations, where all signals are merged into a coarse global space [10, 14]. Such simplification neglects the fact that human reactions naturally emerge from the interaction between transient affective states and stable personal traits. Consequently, these methods struggle to capture the hierarchical organization of speaker behaviour, in which *fine-grained* and *coarse-grained* cues jointly shape observable responses; and ***Problem II:*** *Controllable and Multiple Appropriate Reactions Generation*. Existing models typically achieve facial reaction diversity by sampling from latent distribution spaces [29,43,52], without explicitly modeling the underlying cognitive factors that drive such variation. However, humans can produce different yet contextually appropriate reactions to the same behaviour due to selective attention and cognitive appraisal. As a result, the diversity generated by current methods often lacks psychological plausibility and controllability, limiting their ability to produce meaningful variations aligned with human perception.

In this paper, we propose MREACTOR, a novel offline MAFRG framework that produces multiple diverse yet contextually AFRs from each speaker behaviour. Unlike prior *online* MAFRG systems [7,32,43] that generate AFRs frame-by-frame based only on current and past speaker behaviour, our MREACTOR operates in an *offline* manner by processing the entire speaker sequence beforehand, thereby leveraging future speaker behaviour for more accurate intent and emotional inference.

Inspired by the Human Model Processor (HMP) [3, 17] that models human behaviour response as a cognitive loop of *perceiving*, *interpreting*, and *acting*, our MREACTOR follows the same loop, where: 1) the **Hierarchical Cognition Disentanglement (HCD)** module addresses **Problem I** by disentangling speaker behaviour into *fine-grained* and *coarse-grained* traits, enabling hierarchical cognition interpretation instead of collapsing all factors into a single coarse latent; and 2) The **Multiple AFR Generation (MAG)** module addresses the **Problem II** by generating cognitively meaningful diversity, where multiple reactions are produced by conditioning on contextual behaviour and a subject-aware affect dictionary rather than relying on random latent sampling.

Our main contributions are summarized as follows:

- We propose MREACTOR, the novel *offline* MAG framework that integrates audio, text, and visual cues for comprehensive speaker behaviour understanding, enabling more accurate and contextually appropriate generation of multiple AFRs.
- We introduce the *HCD* module that separates fine-grained dynamics from coarse speaker behaviour cues. With *relative attention* and *subject attention*, our *MAG* module further enables semantically coherent and controllable multiple AFRs.
- Extensive experiments on the MARS dataset demonstrate that MREACTOR surpasses prior methods by 44.4% on FRCorr (0.780), increases diversity (FRDiv 0.2931, FRVar 0.1507), and achieves the lowest synchrony FRSyn (46.89), producing natural, coherent, and semantically aligned facial reactions.

2. Related Work

Existing FRG methods can be divided into *one-to-one mapping* approaches and *one-to-many mapping* approaches.

Facial Reaction Generation (FRG). Early *one-to-one mapping* facial reaction generation approaches were typically developed with the objective to reproduce the paired real facial reaction from each input speaker behaviour [23, 26]. For instance, DyadGAN [8] employed a two-stage conditional GAN to generate AFRs from the speaker's facial action units. Later works extended this idea by incorporating richer speaker behaviour cues such as audio, language, and facial features [9, 20, 22, 30, 35, 49], while some approaches further utilised Neural Architecture Search to enhance the generated facial reactions [30, 49]. Since FRG is naturally a *one-to-many mapping* task [34], deterministic approaches often face the ill-posed *one-to-many mapping* problem during training, causing the model to converge toward generating incorrect average facial reactions.

Multiple Appropriate Facial Reaction Generation (MAFRG). The MAFRG task [34] aims to generate diverse yet appropriate facial reactions in response to each speaker's behaviour, addressing the ill-posed *one-to-many* *mapping* problem. Early MAFRG methods primarily relied on distribution-based modeling: REGNN [43] employed reversible graph neural networks with Gaussian distributions, ReactFace [13] extended this to online scenarios, while Trans-GMM [21] utilised VAE architectures for flexible distribution learning. Transformer-based approaches such as UniFormer [10] and BEAMER [7] integrated multi-modal cues through unified architectures to align speaker behaviours and reactions. However, these methods rely on single-level latent representations [10,14] and random latent sampling, failing to capture hierarchical speaker behaviours or provide cognitively grounded controllability.

Recent advances have explored diffusion models and personalisation strategies. FRDiff [46] and SC-Diff [16] apply scattering-conditioned diffusion, EAT-Face [39] and EmoTalker [29] incorporate emotion-controllable frameworks, while the REACT2024 winner [4] advanced online prediction using Finite Scalar Quantisation. Personalisation-oriented methods include PerFRDiff [53] with personalised weight editing, PerReactor [52] disentangling listener style, and hierarchical decoupling-fusion framework [14] for multi-modal coherent synthesis. Despite these advances, existing methods lack explicit mechanisms for hierarchical cognition interpretation (separating fine-grained affective states from coarse-grained traits) and controllable diversity generation grounded in cognitive principles. In contrast, our MREACTOR addresses both challenges through explicit hierarchical disentanglement and subject-aware affect dictionaries, enabling semantically coherent multiple AFRs aligned with human cognitive processes.

3. Methodology

Hypothesis. Our proposed offline multiple appropriate facial reaction generation approach, MREACTOR, is inspired by the cognitive loop of the Human Model Processor (HMP) [3]. The HMP framework conceptualizes human information processing as a continuous cycle comprising three fundamental stages: *perceiving* multi-modal external stimuli through sensory systems, *interpreting* them cognitively through mental representations, and *acting* through corresponding motor behavioral responses. Critically, Card et al. [3] and Miller [17] establish that human cognition during communication operates hierarchically across multiple temporal scales—where stable, slowly-varying coarse-level patterns provide contextual scaffolding for interpreting rapidly-changing fine-level cues. This hierarchical organization enables humans to generate diverse yet contextually appropriate reactions by dynamically integrating information across different levels of abstraction. Inspired by hierarchical human perception and cognitive architectures, we propose the Hierarchical Cognition Disentanglement (HCD) module. HCD disentangles input behaviors into fine-grained, low-level local actions and coarse-grained, high-level contextual patterns, which respectively capture immediate affective cues and long-term emotional intent. This hierarchical integration allows the model to mirror human cognition, generating multiple AFRs with superior coherence and identity consistency.

3.1. Overview

Given a multi-modal speaker behaviour segment $B_S^{1:T} = \{A_S^{1:T}, L_S^{1:T}, F_S^{1:T}\}$, with $A_S^{1:T}$, $L_S^{1:T}$, and $F_S^{1:T}$ representing the audio, linguistic, and visual streams, respectively, the main goal of our offline MREACTOR is to generate N multiple AFRs by considering the entire multi-modal speaker behaviour sequence $B_S^{1:T}$ as:

$$P_L^{1:T} = \text{MREACTOR}(B_S^{1:T}), \tag{1}$$

where $P_L^{1:T} = \{p_L(B_S^{1:T})_1, \cdots, p_L(B_S^{1:T})_N\}$, and $p_L(B_S^{1:T})_n$ is the n-th AFRs consisting of T facial frames. Although the response sequence may not necessarily have the same length as the input, here we uniformly use T to denote the sequence length for simplicity. Since MREACTOR conditions on the entire speaker behaviour segment $B_S^{1:T}$, it integrates both preceding ($B_S^{1:T-1}$), current (B_S^t) and subsequent ($B_S^{t+1:T}$) speaker behaviours when predicting and generating each AFRs frame $P_L^{1:T}$. This highlights the key distinction between our offline approach and the online ones, which rely solely on past frames. Our full-context offline approach also mirrors the cognitive mechanism described by the HMP [3], where the cognitive processor evaluates the *entire perceptual input before responding*, thereby producing facial reactions that are globally coherent and more consistent to natural human behaviour.

An overview of our proposed MREACTOR is illustrated in Figure 2. It comprises three core modules in accordance to the HMP hypothesis: the **Multi-modal Behaviour Perception (MBP)** module, the **Hierarchical Cognition Disentanglement (HCD)** module, and the **Multiple AFR Generation (MAG)** module. The **MBP** first extracts the speaker's audio (e.g., prosody, pitch, and speaking dynamics that are strongly correlated with emotional intensity and conversational timing), linguistic (e.g., semantic contexts), and facial behaviour (e.g., expression coefficients and head poses reflecting nuanced social signals) features using Enc_A (Wav2Vec 2.0 [1]), Enc_L (RoBERTa [11]), and Enc_F (FaceVerse [40]), respectively, as these play an essential role in triggering human behaviour responses. To address temporal asynchrony across modalities (e.g., delayed alignment between speech and facial expressions) and handle potentially conflicting signals (e.g., contradictions between prosody and semantic content), MBP employs a temporal correlation learning module [48] that captures both intra- and inter-modal dependencies. This module dynam-

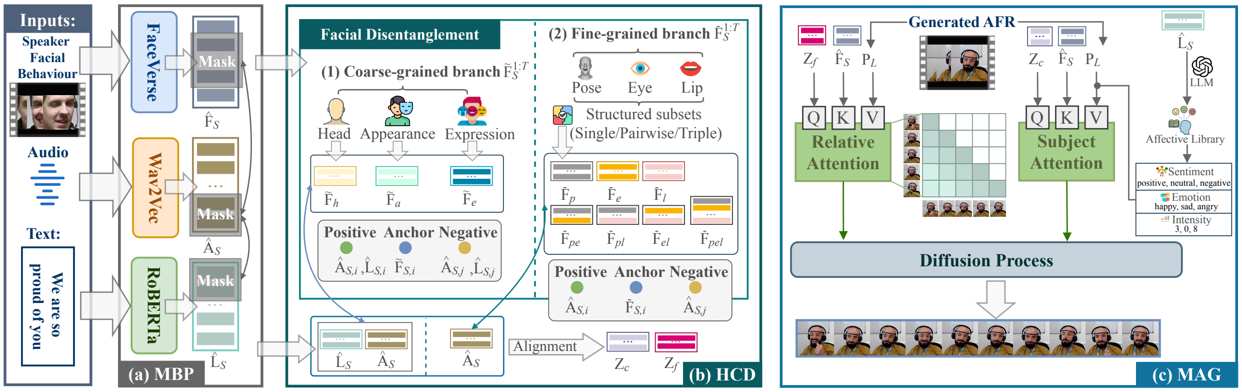


Figure 2. **The pipeline of the proposed MREACTOR. (a) MBP:** Extracts multi-modal audio, linguistic, and facial cues $\hat{\mathbf{A}}_S^{1:T}$, $\hat{\mathbf{L}}_S^{1:T}$, and $\hat{\mathbf{F}}_S^{1:T}$. **(b) HCD:** Disentangles the speaker's facial behaviour into *coarse-grained* (stable appearance, head pose, expression) and *fine-grained* (eye, lip, and structured pose dynamics) branches, and aligns verbal and non-verbal cues to obtain $\mathbf{Z}_c^{1:T}$ and $\mathbf{Z}_f^{1:T}$. **(c) MAG:** Fuses *coarse-grained* and *fine-grained* representations through *relative attention* and *subject attention*, and employs a diffusion model to generate multiple coherent and diverse AFRs. For clarity, the explicit temporal index $\{1{:}\mathrm{T}\}$ is omitted for most variables in the diagram. Unless otherwise specified, all intermediate representations denote temporal sequences.

ically identifies frames with salient behavioral cues while applying temporal masking [48] to suppress segments with weak cross-modal support or conflicting signals, thereby enforcing temporal alignment and reducing noise. These can be formulated as:

$$\hat{\mathbf{A}}_S^{1:T}, \hat{\mathbf{L}}_S^{1:T}, \hat{\mathbf{F}}_S^{1:T} = \mathrm{MBP}\big(A_S^{1:T}, L_S^{1:T}, F_S^{1:T}\big), \quad (2)$$

where $\mathrm{MBP} = \{\mathrm{Enc_A}, \mathrm{Enc_L}, \mathrm{Enc_F}\}$, $\hat{\mathbf{A}}_S^{1:T}$, $\hat{\mathbf{L}}_S^{1:T}$ and $\hat{\mathbf{F}}_S^{1:T}$ represent the extracted speaker audio, linguistic, and facial feature embeddings, respectively. Full details of the temporal correlation learning and masking mechanism are provided in the supplementary material.

The **HCD** module performs reasoning over the encoded speaker behaviours $\hat{\mathbf{A}}_S^{1:T}$ (audio), $\hat{\mathbf{L}}_S^{1:T}$ (text), and $\hat{\mathbf{F}}_S^{1:T}$ (face). Facial behaviour contains two distinct temporal patterns: rapid local motions (lip, eye, pose) and stable global attributes (head pose, appearance, expression). Directly merging these heterogeneous patterns causes transient motions to dominate the representation and weaken the modelling of long-term context. Therefore, HCD separates facial features into a *fine-grained* branch for transient expressions and a *coarse-grained* branch for stable cues. Audio and text remain single-stream because their encoders already perform temporal aggregation, so additional splitting would be redundant and unnecessary. The is formulated as:

$$(\mathbf{Z}_f^{1:T}, \mathbf{Z}_c^{1:T}) = \mathrm{HCD}(\hat{\mathbf{A}}_S^{1:T}, \hat{\mathbf{L}}_S^{1:T}, \hat{\mathbf{F}}_S^{1:T}), \quad (3)$$

where $\mathbf{Z}_f^{1:T}$ and $\mathbf{Z}_c^{1:T}$ represent the speaker fine-grained and croase-grained features, respectively.

Finally, the **MAG** module generates multiple listener facial reactions from the disentangled fine-grained and coarse-grained speaker behaviour representations $(\mathbf{Z}_f^{1:T}, \mathbf{Z}_c^{1:T})$. It employs a *contextual behaviour* sub-module and an *expressive recovery* sub-module that attend to fine-grained cues in $\mathbf{Z}_f^{1:T}$ and adaptively modulate the coarse-grained context in $\mathbf{Z}_c^{1:T}$. MAG ensures that the generated AFRs stay coherent with the speaker's behaviour by capturing expression-related dependencies and adapting to contextual cues. The process is formulated as:

$$\mathbf{P}_L^{1:T} = \mathrm{MAG}(\mathbf{Z}_f^{1:T}, \mathbf{Z}_c^{1:T}, L_S^{1:T}), \quad (4)$$

where $\mathbf{P}_L^{1:T}$ denotes the generated listener's multiple AFRs.

In the following, we provide detailed descriptions of the major modules (**HCD** and **MAG**) in our framework.

3.2. Hierarchical Cognition Disentanglement

The **HCD** module models how a listener cognitively interprets multi-level behavioural cues from the speaker via audio $\hat{\mathbf{A}}_S^{1:T}$, linguistic $\hat{\mathbf{L}}_S^{1:T}$, and facial $\hat{\mathbf{F}}_S^{1:T}$ representations. Unlike audio and text, whose encoders already aggregate information across temporal ranges, facial behaviour involves two structurally different temporal patterns: local attributes (lip motion, eye blinks, or subtle pose changes) and global tendencies (head orientation, appearance, or expression style). Combining these patterns in a single embedding mixes short-term fluctuations with long-term information and weakens the overall temporal clarity. Therefore, HCD disentangles speaker facial representation $\hat{\mathbf{F}}_S^{1:T}$ into a *fine-grained* and a *coarse-grained* representation, and aligns speaker verbal and non-verbal cues.

Fine-grained Facial Disentanglement. Local facial cues are extracted by sampling regions around facial landmarks, yielding $\overline{F}_S^{1:T} = \{\overline{F}_l^{1:T}, \overline{F}_e^{1:T}, \overline{F}_p^{1:T}, \overline{F}_{le}^{1:T}, \overline{F}_{lp}^{1:T}, \overline{F}_{ep}^{1:T}, \overline{F}_{lep}^{1:T}\}$. These structured subsets, including single cues (e.g., *lip* [50], *eye* [45], *pose* [42]), their pairwise combinations, and the full triplet, enable the model to more effectively capture fine-grained local speaker micro-expressions. Positive/negative speaker verbal audio and non-verbal face cues alignment pairs are defined as:

$$\mathcal{P} = \{(\overline{\mathbf{F}}_{S,i}^{1:T}, \hat{\mathbf{A}}_{S,i}^{1:T})\}, \quad \mathcal{N} = \{(\overline{\mathbf{F}}_{S,i}^{1:T}, \hat{\mathbf{A}}_{S,j}^{1:T}) \mid j \neq i\}, \tag{5}$$

and the fine-grained representation is computed as:

$$z_f^t = \alpha\big(\langle \overline{\mathbf{F}}_{S,i}^t, \hat{\mathbf{A}}_{S,i}^t \rangle - \beta\big), \quad t = 1, \ldots, T, \tag{6}$$

$$\mathbf{Z}_f^{1:T} = [\, z_f^1, z_f^2, \ldots, z_f^T \,], \tag{7}$$

where $\langle \cdot, \cdot \rangle$ denotes cosine similarity [28], and α and β are the scaling and margin hyperparameters, respectively, while i and j index different samples.

Coarse-grained Facial Disentanglement. Stable expressive patterns are modeled by globally sampling speaker face cues from the MPB representation as $\tilde{\mathbf{F}}_S^{1:T} = \{F_h^{1:T}, F_a^{1:T}, F_e^{1:T}\}$. Aligning global speaker facial cues, including head [2], appearance [47], and expression [36], with audio and linguistic cues enables the model to learn expressive consistency across modalities.

$$\begin{aligned} \mathcal{P} &= \{(\tilde{\mathbf{F}}_{S,i}^{1:T}, (\hat{\mathbf{A}}_{S,i}^{1:T}, \hat{\mathbf{L}}_{S,i}^{1:T}))\}, \\ \mathcal{N} &= \{(\tilde{\mathbf{F}}_{S,i}^{1:T}, (\hat{\mathbf{A}}_{S,j}^{1:T}, \hat{\mathbf{L}}_{S,j}^{1:T})) \mid j \neq i\}, \end{aligned} \tag{8}$$

while the coarse-grained facial disentanglement is computed as:

$$z_c^t = \gamma\Big(\mathrm{W}_2\big(\tilde{\mathbf{F}}_{S,i}^t, (\hat{\mathbf{A}}_{S,i}^t, \hat{\mathbf{L}}_{S,i}^t)\big) - \delta\Big), \quad t = 1, \ldots, T, \tag{9}$$

$$\mathbf{Z}_c^{1:T} = [\, z_c^1, z_c^2, \ldots, z_c^T \,], \tag{10}$$

where $\mathrm{W}_2(\cdot)$ denotes the 2-Wasserstein distance [38], and γ and δ are the scaling and margin hyperparameters. To handle the differing characteristics of the two cue types, we apply cosine similarity for fine-grained cues and the 2-Wasserstein distance for coarse-grained cues. Cosine similarity emphasizes angular differences, making it well suited for fine-grained cues whose expressive variations are highly directional and subtle. In contrast, coarse-grained cues reflect more distributionally stable behaviours over time; therefore, the 2-Wasserstein distance better aligns their cross-modal distribution geometry. Our HCD consequently produces interpretable multi-level facial embeddings, where fine-grained features capture rapid expressive changes and coarse-grained features encode stable behavioural context, yielding multiple yet consistent AFRs.

3.3. Multiple AFR Generation

The MAG module generates multiple AFRs by modelling how listeners integrate contextual cues, sentiment, and emotion in response to a speaker's behaviour. It consists of two key sub-modules: *contextual behaviour* and *expressive recovery*. The contextual behaviour sub-module learns speaker-specific behavioural cues and models how these cues influence the temporal evolution of multiple AFRs [25]. In parallel, the expressive recovery sub-module generates AFRs that reflect sentiment, emotion, and reaction intensity based on the relevant subject. To ensure that the generated reactions remain both contextually coherent and expressively consistent, MAG combines the outputs of these sub-modules using our proposed *relative attention* and *subject attention*.

Contextual Behaviour. Ensuring temporally coherent AFRs that remain sensitive to the speaker's evolving cues requires a contextual behaviour module that models how past speaker states influence the current reaction. Standard self-attention computes attention weights solely from content similarity, i.e., the dot product $\langle q_x, k_y \rangle$ [37]. In long sequences, however, this may lead to attention drift, where temporally distant yet coincidentally similar frames receive disproportionately high weights. MAG incorporates a *relative attention* term into the attention logits to explicitly regulate how historical cues influence the current moment as:

$$\mathbf{R} = \begin{cases} \epsilon(x-y) + r, & y \leq x, \\ -\infty, & y > x, \end{cases} \tag{11}$$

where $\epsilon(\cdot)$ is a learnable distance coefficient determining how influence decays with temporal distance and r is a learnable bias; $y > x$ is masked to strictly prevent future information leakage. The *contextual behaviour* sub-module is computed as:

$$\mathcal{C} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_c \mathbf{K}_c^\top}{\tau_c} + \mathbf{R}\right) \mathbf{V}_c, \tag{12}$$

where $\mathbf{Q}_c = \mathbf{Z}_f^{1:T} \mathbf{W}_c^Q$, $\mathbf{K}_c = \hat{\mathbf{F}}_S^{1:T} \mathbf{W}_c^K$, and $\mathbf{V}_c = \mathbf{P}_L^{1:T} \mathbf{W}_c^V$. The matrices $\mathbf{W}_c^Q$, $\mathbf{W}_c^K$, and $\mathbf{W}_c^V$ are trainable linear projections that transform the input features into the query, key, and value representations required for attention computation. Adding $\mathbf{R}$ introduces a relative-position bias into the attention logits:

$$\alpha_{xy} \propto \exp(\langle q_x, k_y \rangle / \tau_c + \epsilon(x-y) + r)\,. \tag{13}$$

The term $\epsilon(x-y)$ reflects the temporal distance between frame x and y, decreasing the logit as $|x-y|$ grows. This prevents distant but coincidentally similar frames from dominating the attention. By incorporating relative positional bias alongside content similarity, the model focuses on contextually relevant recent cues and produces more temporally consistent reactions.

Expressive Recovery. Generating multiple AFRs is challenging because standard attention mechanisms determine only *which* cues to attend to, but provide little control over *how* the reaction should be expressed, such as its sentiment polarity, emotion category, or intensity. Existing methods based on fixed expressive styles [12, 16, 32] implicitly constrain the subject to a single expression mode, which limits flexibility when affect changes dynamically over time. To address this limitation, MAG introduces a *subject-aware attention* mechanism that modulates the value space using an explicit affective prior:

$$\mathcal{E}^t = \mathrm{Softmax}\left(\frac{\mathbf{Q}_e^t(\mathbf{K}_e^t)^\top}{\tau_e}\right)\mathbf{S}^t\mathbf{V}_e^t, \tag{14}$$

where $\mathbf{Q}_e^t = \mathbf{Z}_c^t\mathbf{W}_e^Q$, $\mathbf{K}_e^t = \tilde{\mathbf{F}}_S^t\mathbf{W}_e^K$, and $\mathbf{V}_e^t = \mathbf{P}_L^t\mathbf{W}_e^V$. Unlike prior work that applies a single static affective prior to the whole sequence, our $\mathbf{S}^t$ is computed at each timestep t. This design allows the model to capture affective variation within a single utterance as emotions shift during its unfolding, while such variations also accumulate across multiple conversational turns, enabling explicit modeling of affective dynamics throughout the dialogue.

Dynamic and Interpretable Subject-Aware Affect Dictionary. The time-varying subject prior $\mathbf{S}^{1:T}$ is constructed through a three-stage pipeline designed to balance semantic richness with interpretability. At each frame t, we use GPT-4 only to generate candidate affective subjects conditioned on the current speaker linguistic representation $\hat{L}_S^t$:

$$\mathcal{T}_{\text{raw}}^t = \{\textit{achievement}, \textit{acceptance}, \ldots\}.$$

Importantly, the LLM is used solely as a candidate generator and is *not* involved in any subsequent scoring, weighting, or decision-making step. To remove black-box ambiguity and mitigate hallucination risks, all candidate subjects are passed through a deterministic lexicon-based validation procedure. Specifically, a candidate is retained only if it is supported by multiple psychology-informed affective lexicons with consistent affective evidence, yielding the validated set $\mathcal{T}_{\text{validated}}^t$. In our implementation, this validation step relies on resources such as ANEW, VADER, and SentiWordNet [18, 24, 41]. Therefore, all retained subjects are grounded in established affective lexicons rather than relying directly on LLM outputs.

For each validated subject $l_i^t \in \mathcal{T}_{\text{validated}}^t$, we retrieve its sentiment polarity s_i^t and emotion intensity $\phi(e_i^t)$ directly from the validated lexicons. The frame-level affective prior is then explicitly computed as

$$\mathbf{S}^t = \frac{1}{Z^t}\sum_{l_i^t \in \mathcal{T}_{\text{validated}}^t} s_i^t \cdot \phi(e_i^t), \tag{15}$$

where Z^t is a normalization factor at timestep t. As a result, $\mathbf{S}^t$ is fully determined by lexicon-derived sentiment and emotion signals, making the expressive prior transparent and reproducible. By computing $\mathbf{S}^{1:T}$ dynamically instead of using a fixed global prior, the model can track fluctuating sentiment and emotion intensity as speaker content evolves, thereby producing more realistic and psychologically grounded listener reactions.

4. Experiments

4.1. Experimental Settings

Dataset. Our MREACTOR is evaluated on a publicly available audio-visual dyadic interaction dataset provided by the REACT2025 challenge[1] [31]. The dataset consists of 137 human-human dyadic interaction multi-modal clips recorded from 23 speakers and 137 listeners. Specifically, each clip captures the behaviour of a human speaker and listener in separate files, resulting in a total of 270 multi-modal clips, with durations ranging from 20 to 35 minutes. All clips are split into 2856 multi-modal session pairs.

Implementation Details. We follow the preprocessing pipeline of the REGNN base system [43]. All facial frames are cropped and resized to 224×224 pixels. We employ the FaceVerse model to extract 3D Morphable Model (3DMM) coefficients and utilize PIRender to reconstruct the corresponding facial frame sequences from the extracted parameters. Our implementation is based on PyTorch and runs on a single NVIDIA Tesla A800 GPU with 80 GB of memory for a total of 150 training epochs. We train the PerFRDiff baseline, Unifarn [10], SC-Diff [16], and our proposed MREACTOR model on the REACT2025 dataset [31]. Training is performed using the AdamW optimizer with an initial learning rate of 1×10^{-4}, a batch size of 4, and a cosine annealing learning rate schedule. During evaluation, the batch size is reduced to 1 to enable frame-wise inference.

Metrics. Consistent with previous challenges [31–33], we evaluate the generated facial reaction attributes across four key dimensions: (i) *Appropriateness* (measured by FRCorr and FRdist), (ii) *Diversity* (measured by FRDiv, FRDvs, and FRVar, where FRDiv quantifies the diversity of multiple generated AFRs in response to the same speaker behaviour), and (iii) *Synchrony* (FRSyn). For further details, please refer to [34].

4.2. Method Comparison

Table 1 presents a comprehensive quantitative comparison between our proposed MREACTOR and existing MAFRG approaches on the MARS test set. It can be clearly observed that MREACTOR achieves superior performance across all evaluation metrics, demonstrating its effectiveness in generating multiple AFRs. In terms of

[1] https://sites.google.com/view/react2025

Table 1. Baseline results achieved on the test set of the MARS dataset, where B_Random randomly samples $\alpha = 10$ facial reaction sequences from a Gaussian distribution; B_Mime generates facial reactions by mimicking the corresponding speaker facial behaviours; B_MeanSeq and B_MeanFr generate facial reactions by averaging the sequence- and frame-wise facial reactions in the training set, respectively. For a fair comparison, we primarily benchmark against offline baselines that match our setting (PerReactor [52], SC-Diff [16], REGNN [43], Trans-VAE [32]). USTC-AC [12] and UniFaRN [10] support both online and offline modes, while PerFRDiff [53] is an online method; these are reported only for reference following common practice.

Method	Appropriateness		Diversity		Synchrony
	FRCorr (↑)	FRDist (↓)	FRDiv (↑)	FRVar (↑)	FRSyn (↓)
GT	1	0.00	0.1876	0.0669	48.66
B_Random	0.03	474.68	0.3342	0.1671	46.64
B_Mime	0.52	206.02	0.0000	0.0766	43.70
B_MeanFr	0.00	205.65	0.0000	0.0000	49.00
Offline Results					
Trans-VAE [32]	0.30	181.72	0.0076	0.0083	49.00
Unifarn [10]	0.55	225.63	0.1538	0.0815	48.36
REGNN [43]	0.54	155.49	0.0012	0.0048	44.81
USTC-AC [12]	0.53	229.12	0.1451	0.0768	48.95
PerFRDiff [53]	0.58	217.32	0.1596	0.0862	48.66
PerReactor [52]	0.54	226.89	0.1513	0.0792	48.57
SC-Diff [16]	0.58	214.67	0.0402	0.0811	49.00
Ours	**0.780**	**165.26**	**0.2931**	**0.1507**	**46.89**

appropriateness, MREACTOR attains the highest FRCorr score of 0.780, yielding relative improvements of 44.4% and 41.8% compared to the previous state-of-the-art methods PerFRDiff [53] (0.58) and SC-Diff [16] (0.58), respectively. Importantly, although FRCorr evaluates appropriateness, strong performance on this metric should not be interpreted as evidence of deterministic mapping. Instead, it indicates that each of our diverse generated reactions is contextually appropriate, effectively addressing the one-to-many challenge.

Furthermore, MREACTOR consistently outperforms all competing methods in diversity and realism metrics. Specifically, it achieves an FRDiv of 0.2931, surpassing Unifarn [10] (0.1538) and PerFRDiff [53] (0.1596) by 90.6% and 83.7%, respectively. For FRVar, our model attains 0.1507, outperforming REGNN [43] (0.0048) and SC-Diff [16] (0.0811) by over 31× and 85.8%, respectively. Regarding temporal consistency, MREACTOR achieves the lowest FRSyn value of 46.89, outperforming Unifarn (48.36) and PerReactor [52] (48.57), indicating smoother and more human-like transition dynamics. As shown in Table 1, all baseline models outperform the naive B_Random and B_MeanFr settings, indicating their ability to generate non-trivial AFRs. The B_Mime baseline achieves a comparatively high FRCorr score, which aligns with the well-established facial mimicry effect [5]. In contrast, MREACTOR goes beyond surface-level mimicry by producing personalized and semantically coherent reactions that remain temporally stable while preserving expressive diversity. Moreover, Fig. 3 further demonstrates that our *MReactor* generates a richer set of plausible and varied generic AFRs than existing baselines.

Table 2. Ablation study of the MBP, HCD, and MAG modules in MREACTOR on the MARS test set. REGNN is the base system [43]. 1) Base system + HCD + MAG removes the MBP module while retaining HCD and MAG. 2) Base system + MBP + MAG removes the HCD module while retaining MBP and MAG. 3) Base system + MBP + HCD removes the MAG module while retaining MBP and HCD.

Method	Appropriateness		Diversity		Synchrony
	FRCorr (↑)	FRDist (↓)	FRDiv (↑)	FRVar (↑)	FRSyn (↓)
base + HCD+ MAG	0.623	172.32	0.1738	0.1329	47.65
base + MBP + MAG	0.507	192.07	0.1129	0.0796	48.03
base + MBP + HCD	0.439	206.95	0.0867	0.0583	48.75
Ours	**0.780**	**165.26**	**0.2931**	**0.1507**	**46.89**

Table 3. Modality ablation results on the MARS test set, evaluating the contribution of each single modality (audio, text, video) compared with the full multi-modal MREACTOR.

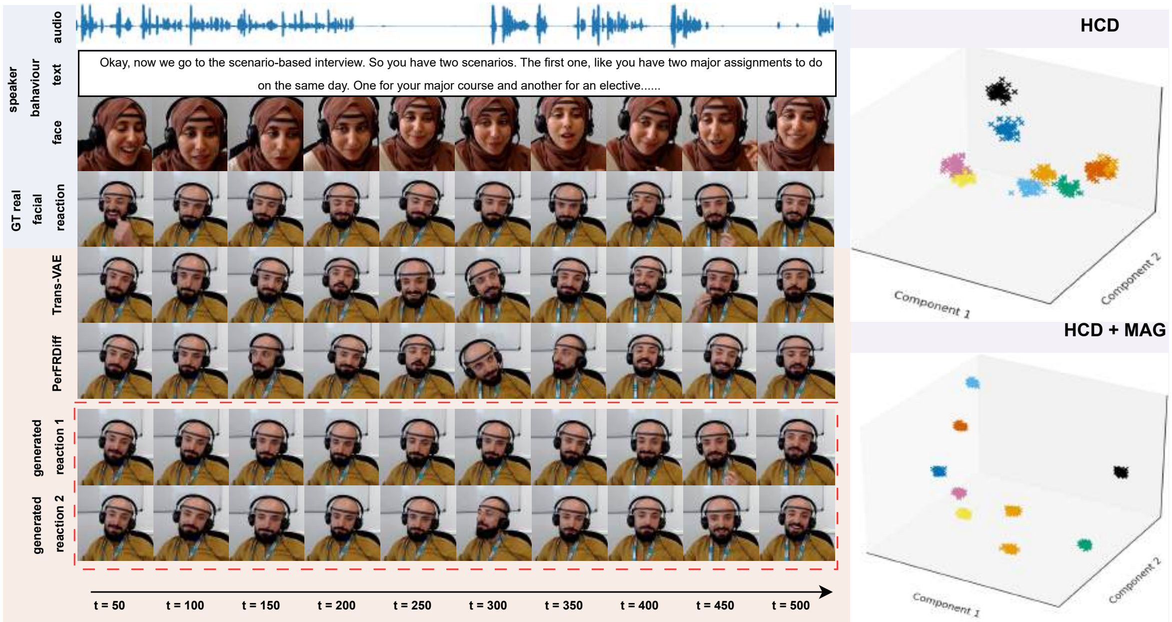
Method	Appropriateness		Diversity		Synchrony
	FRCorr (↑)	FRDist (↓)	FRDiv (↑)	FRVar (↑)	FRSyn (↓)
Audio	0.56	186.29	0.1567	0.0912	48.65
Text	0.45	203.32	0.0898	0.0607	49.03
Video	0.65	170.21	0.1786	0.1393	47.73
Ours	**0.78**	**165.26**	**0.2931**	**0.1507**	**46.89**

4.3. Ablation Study

Tables 2 and 3 evaluate the contribution of (i) each core module in MREACTOR (MBP, HCD, and MAG) and (ii) each input modality (audio, text, and video), allowing us to assess how perception, cognition, and generation components influence performance.

Contributions of Core Modules. Table 2 shows that MBP ensures robust perception (FRCorr = 0.623), HCD enhances hierarchical reasoning (FRCorr = 0.507), and MAG alone yields weaker results due to limited contextual cues (FRCorr = 0.439). Integrating all modules brings clear gains across all metrics (FRCorr = 0.780, FRDiv = 0.2931, FRVar = 0.1507, FRSyn = 46.89), verifying that perception, cognition, and generation work synergistically to form a cognitively grounded perceive–interpret–act loop.

Contributions of Different Modalities. As shown in Table 3, all modalities contribute to generating appropriate facial reactions. Video performs best among single modalities (FRCorr = 0.65), highlighting the importance of vi-



(a) visualisation of generated facial reactions

(b) t-SNE visualisation

Figure 3. (a) Visualizations of facial reactions generated by different approaches, where our method produces noticeably richer head movements and more diverse facial expressions. (b) t-SNE visualizations of features from HCD alone and from HCD combined with MAG in our MREACTOR.

sual cues, while audio and text also provide complementary cues in dynamic human-human interactions. The full multi-modal MREACTOR achieves the best overall performance (FRCorr $=$ 0.78, FRDist $=$ 165.26, FRDiv $=$ 0.2931, FRVar $=$ 0.1507, FRSyn $=$ 46.89), confirming the effectiveness of joint verbal (audio, text) and nonverbal features (video) fusion and validating our full-context offline design.

4.4. Attention Analysis

Table 4 evaluates the role of different attention mechanisms in MREACTOR. Compared to SA or CA alone, our integrated *relative attention* (contextual behaviour) and *subject attention* (expressive recovery) sub-modules achieve substantially higher appropriateness (FRCorr $=$ 0.78) and diversity (FRDiv $=$ 0.2931, FRVar $=$ 0.1507), while improving synchrony (FRSyn $=$ 46.89). The *relative attention* captures temporal dependencies by prioritizing contextually relevant cues and suppressing spurious correlations from distant frames, while *subject attention* injects personalized expressiveness by modulating the value space via sentiment strength and emotion intensity. Their joint use yields coherent, controllable, and diverse AFRs.

5. Conclusion

In this paper, we introduced MREACTOR, the novel *offline* and cognitively grounded framework for Multiple Appropriate Facial Reaction Generation (MAFRG). Drawing inspiration from the Human Model Processor, MREACTOR unifies perception, cognition, and action through three coordinated modules: *MBP* for robust cue extraction, *HCD* for structured local and global reasoning, and *MAG* for personalized and semantically coherent reaction synthesis. Extensive experiments on the MARS dataset demonstrate that MREACTOR achieves state-of-the-art performance, surpassing previous methods by 44.4% in appropriateness (FRCorr 0.780) and significantly improving diversity (FRDiv 0.2931, FRVar 0.1507) with the lowest synchrony FRSyn (46.89). These results confirm that MREACTOR enables natural, diverse, and cognitively consistent facial reactions, offering a promising step toward more empathetic human–computer interaction.

Table 4. Impact of different attention mechanisms on MREACTOR. SA and CA denote the *self-attention and cross-attention* modules in [53], respectively.

Method	Appropriateness		Diversity		Synchrony
	FRCorr (↑)	FRDist (↓)	FRDiv (↑)	FRVar (↑)	FRSyn (↓)
SA	0.71	178.83	0.1879	0.1423	47.17
CA	0.72	179.98	0.1891	0.1457	47.05
SA + CA	0.68	175.29	0.1783	0.1409	47.32
Ours	**0.78**	**165.26**	**0.2931**	**0.1507**	**46.89**

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